Movie Genre Data Analysis

This notebook focuses on analyzing movie genres using the IMDb movies dataset. We will explore various aspects of genres, including their frequency, budget, revenue, popularity, and voting averages.

Research Questions

1. Which genres are the most common (number of movies made)?
2. Which genres have high average budget and revenue?
3. Which genres have high average popularity?
4. Which genres have the highest number of movies with a voting average >= 8?

Research Hypotheses

1. The best movies according to vote average return high profit and revenue.
2. The best movies according to popularity return high profit and revenue.
3. Highly budgeted movies return high revenue and profit.
4. Highly budgeted movies have high popularity.

Acknowledgements

This analysis is based on the IMDb movies dataset provided by AnalystBuilder's Pandas for Data Analysis course. The CSV file used is 'imdb_movies.csv'.

In [ ]:
```
# Importing Pandas Library
import pandas as pd
```

In [ ]:
```
# Reading the CSV file
df = pd.read_csv('imdb_movies.csv')
```

In [ ]:
```
# Displaying the first 5 rows of the dataframe
pd.set_option('display.max_columns', None)
pd.set_option('display.expand_frame_repr', False)

# Displaying the first 5 rows of the dataframe
df.head()
```

Out[ ]:

	id	imdb_id	popularity	budget	revenue	original_title	cast	homepage
0	135397	tt0369610	32.985763	150000000	1513528810	Jurassic World	Chris Pratt|Bryce Dallas Howard|Irrfan Khan|Vi...	http://www.jurassicworld.com/
1	76341	tt1392190	28.419936	150000000	378436354	Mad Max: Fury Road	Tom Hardy|Charlize Theron|Hugh Keays-Byrne|Nic...	http://www.madmaxmovie.com/
2	262500	tt2908446	13.112507	110000000	295238201	Insurgent	Shailene Woodley|Theo James|Kate Winslet|Ansel...	http://www.thedivergentseries.movie/#insurgent
3	140607	tt2488496	11.173104	200000000	2068178225	Star Wars: The Force Awakens	Harrison Ford|Mark Hamill|Carrie Fisher|Adam D...	http://www.starwars.com/films/star-wars-episod...
4	168259	tt2820852	9.335014	190000000	1506249360	Furious 7	Vin Diesel|Paul Walker|Jason Statham|Michelle ...	http://www.furious7.com/

In [ ]:
```
# Displaying the last 5 rows of the dataframe
```

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 10866 entries, 0 to 10865
Data columns (total 21 columns):
 #   Column                Non-Null Count  Dtype  
---  ------                --------------  -----  
 0   id                    10866 non-null  int64  
 1   imdb_id               10856 non-null  object 
 2   popularity            10866 non-null  float64
 3   budget                10866 non-null  int64  
 4   revenue               10866 non-null  int64  
 5   original_title        10866 non-null  object 
 6   cast                  10790 non-null  object 
 7   homepage              2936 non-null   object 
 8   director              10822 non-null  object 
 9   tagline               8042 non-null   object 
 10  keywords              9373 non-null   object 
 11  overview              10862 non-null  object 
 12  runtime               10866 non-null  int64  
 13  genres                10843 non-null  object 
 14  production_companies  9836 non-null   object 
 15  release_date          10866 non-null  object 
 16  vote_count            10866 non-null  int64  
 17  vote_average          10866 non-null  float64
 18  release_year          10866 non-null  int64  
 19  budget_adj            10866 non-null  float64
 20  revenue_adj           10866 non-null  float64
dtypes: float64(4), int64(6), object(11)
memory usage: 1.7+ MB
```

In [ ]:
```
# Displaying the last 5 rows of the dataframe
df.describe(include='all')
```

Out[ ]:

	id	imdb_id	popularity	budget	revenue	original_title	cast	homepage
count	10866.000000	10856	10866.000000	1.086600e+04	1.086600e+04	10866	10790	2936
unique	NaN	10855	NaN	NaN	NaN	10571	10719	2896
top	NaN	tt0411951	NaN	NaN	NaN	Hamlet	Louis C.K.	http://www.thehungergames.movie/
freq	NaN	2	NaN	NaN	NaN	4	6	4
mean	66064.177434	NaN	0.646441	1.462570e+07	3.982332e+07	NaN	NaN	NaN
std	92130.136561	NaN	1.000185	3.091321e+07	1.170035e+08	NaN	NaN	NaN
min	5.000000	NaN	0.000065	0.000000e+00	0.000000e+00	NaN	NaN	NaN
25%	10596.250000	NaN	0.207583	0.000000e+00	0.000000e+00	NaN	NaN	NaN
50%	20669.000000	NaN	0.383856	0.000000e+00	0.000000e+00	NaN	NaN	NaN
75%	75610.000000	NaN	0.713817	1.500000e+07	2.400000e+07	NaN	NaN	NaN
max	417859.000000	NaN	32.985763	4.250000e+08	2.781506e+09	NaN	NaN	NaN

In [ ]:
```
# Checking for missing values
df.isnull().sum()
```

Out[ ]:
```
id                        0
imdb_id                  10
popularity                0
budget                    0
revenue                   0
original_title            0
cast                     76
homepage               7930
director                 44
tagline                2824
keywords               1493
overview                  4
runtime                   0
genres                   23
production_companies    1030
release_date              0
vote_count                0
vote_average              0
release_year              0
budget_adj                0
revenue_adj               0
dtype: int64
```

In [ ]:
```
# Find the number of duplicate rows
duplicate_rows = df.duplicated().sum()
print("Number of duplicate rows: ", duplicate_rows)
```

```
Number of duplicate rows:  1
```

Findings So Far

Dataset Overview

- The dataset contains 10,866 entries and 21 columns.
- Columns include `id`, `imdb_id`, `popularity`, `budget`, `revenue`, `original_title`, `cast`, `homepage`, `director`, `tagline`, `keywords`, `overview`, `runtime`, `genres`, `production_companies`, `release_date`, `vote_count`, `vote_average`, `release_year`, `budget_adj`, and `revenue_adj`.

Missing Values

- Some columns have missing values:
  - `imdb_id`: 10 missing
  - `cast`: 76 missing
  - `homepage`: 7930 missing
  - `director`: 44 missing
  - `tagline`: 2824 missing
  - `keywords`: 1493 missing
  - `overview`: 4 missing
  - `genres`: 23 missing
  - `production_companies`: 1030 missing

Duplicate Rows

- There is 1 duplicate row in the dataset.

Descriptive Statistics

- The dataset includes a mix of numerical and categorical data.
- Key statistics for numerical columns:
  - `popularity`: Mean = 0.646, Std = 1.000, Min = 0.000065, Max = 32.985763
  - `budget`: Mean = 14,625,700, Std = 30,913,210, Min = 0, Max = 425,000,000
  - `revenue`: Mean = 39,823,320, Std = 117,003,500, Min = 0, Max = 2,781,506,000
  - `runtime`: Mean = 102.07, Std = 31.38, Min = 0, Max = 900
  - `vote_count`: Mean = 217.39, Std = 575.62, Min = 10, Max = 9767
  - `vote_average`: Mean = 5.97, Std = 0.94, Min = 1.5, Max = 9.2
  - `release_year`: Mean = 2001.32, Std = 12.81, Min = 1960, Max = 2015
  - `budget_adj`: Mean = 17,551,040, Std = 34,306,160, Min = 0, Max = 425,000,000
  - `revenue_adj`: Mean = 51,364,360, Std = 144,632,500, Min = 0, Max = 2,827,124,000

These findings provide a good starting point for further analysis, especially focusing on the research questions and hypotheses related to movie genres.

Research Question 1: Which Genres Are the Most Common (Number of Movies Made)?

Data Cleaning

Handle Missing Values in the `genres` Column

We will fill the missing values in the `genres` column with an empty string to avoid issues during the analysis.

Remove Duplicate Rows

We will remove any duplicate rows from the dataset to ensure the accuracy of our analysis.

In [ ]:
```
# Data Cleaning

# Handle missing values in the `genres` column
df['genres'] = df['genres'].fillna('')

# Remove duplicate rows
df = df.drop_duplicates()
```

Data Preparation

Extract and Count the Occurrences of Each Genre

We will split the `genres` column by the '|' character and count the occurrences of each genre.

In [ ]:
```
# Data Preparation

# Split the genres and count occcurrences
from collections import Counter

# Split genres and flatten the list
genres_list = df['genres'].str.split('|').explode()

# Count occurrences of each genre
genres_count = Counter(genres_list)

# Convert to DataFrame for better visualization
genres_count_df = pd.DataFrame(genres_count.items(), columns=['Genre', 'Count']).sort_values(by='Count', ascend

# Display the top genres
genres_count_df.head()
```

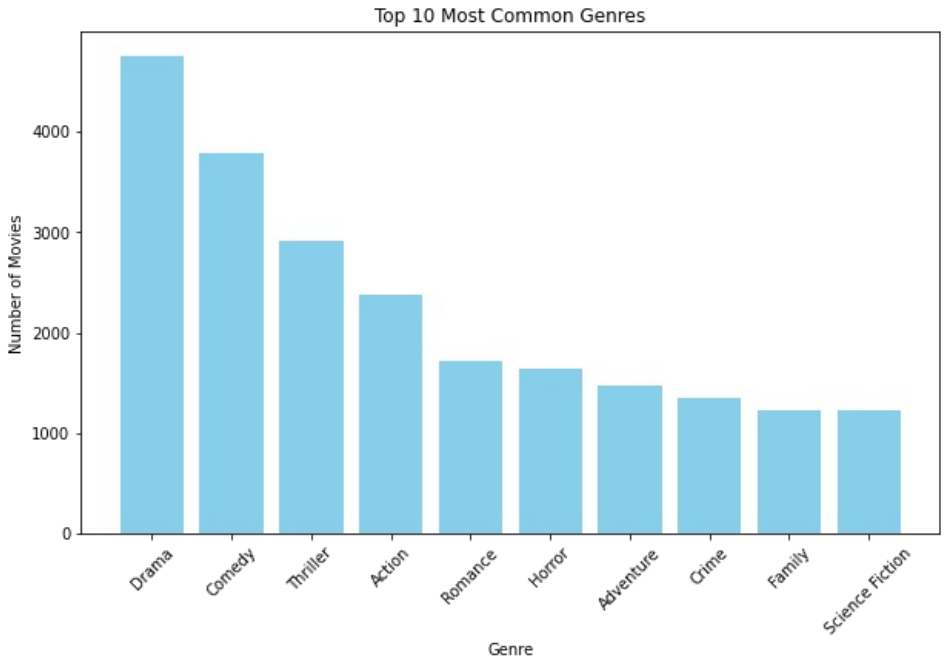
Out[ ]:

	Genre	Count
7	Drama	4760
10	Comedy	3793
3	Thriller	2907
0	Action	2384
12	Romance	1712

Visualization of the Most Common Genres

We will create a bar chart to visualize the distribution of the top genres.

In [ ]:
```
import matplotlib.pyplot as plt
# Plotting the top genres
plt.figure(figsize=(10, 6))
plt.bar(genres_count_df['Genre'].head(10), genres_count_df['Count'].head(10), color='skyblue')
plt.xlabel('Genre')
plt.ylabel('Number of Movies')
plt.title('Top 10 Most Common Genres')
plt.xticks(rotation=45)
plt.show()
```



This bar chart displays the top 10 most common movie genres, ranked by the number of movies in each category. Drama is shown to be the most prevalent genre, followed by Comedy and Thriller, with Science Fiction rounding out the bottom of the list.

Percentage Representation of the Most Common Genres

We will calculate the percentage of movies for each genre relative to the total number of movies.

In [ ]:
```
# Calculate the percentage of movies for each genre
total_movies = len(df)
genres_count_df['Percentage'] = (genres_count_df['Count'] / total_movies) * 100

# Display the top genres with their percentages
genres_count_df.head(10)
```

Out[ ]:

	Genre	Count	Percentage
7	Drama	4760	43.810400
10	Comedy	3793	34.910262
3	Thriller	2907	26.755637
0	Action	2384	21.942016
12	Romance	1712	15.757018
16	Horror	1637	15.066728
1	Adventure	1471	13.538886
5	Crime	1354	12.462034
8	Family	1231	11.329959
2	Science Fiction	1229	11.311551

Analysis of Genre Combinations

We will analyze the most common genre combinations.

In [ ]:
```
# Count occurrences of each genre combination
genre_combinations_count = df['genres'].value_counts().reset_index()
genre_combinations_count.columns = ['Genre Combination', 'Count']

# Display the top genre combinations
genre_combinations_count.head(10)
```

Out[ ]:

	Genre Combination	Count
0	Drama	712
1	Comedy	712
2	Documentary	312
3	Drama|Romance	289
4	Comedy|Drama	280
5	Comedy|Romance	268
6	Horror|Thriller	259
7	Horror	253
8	Comedy|Drama|Romance	222
9	Drama|Thriller	138

Findings for Research Question 1: Which Genres Are the Most Common (Number of Movies Made)?

Top Genres by Count

The most common genres in the dataset are:

1. **Drama**: 4,760 movies (43.81%)
2. **Comedy**: 3,793 movies (34.91%)
3. **Thriller**: 2,907 movies (26.76%)
4. **Action**: 2,384 movies (21.94%)
5. **Romance**: 1,712 movies (15.76%)

Visualization of the Most Common Genres

A bar chart was created to visualize the distribution of the top 10 genres, showing that Drama and Comedy are the most prevalent genres.

Percentage Representation

The percentage of movies for each genre relative to the total number of movies was calculated. Drama and Comedy together account for nearly 80% of the movies in the dataset.

Analysis of Genre Combinations

The most common genre combinations are:

1. **Drama**: 712 movies
2. **Comedy**: 712 movies
3. **Documentary**: 312 movies
4. **Drama|Romance**: 289 movies
5. **Comedy|Drama**: 280 movies

These findings provide a comprehensive understanding of the most common genres and their combinations in the dataset.

Research Question 2: Which Genres Have High Average Budget and Revenue?

To answer this question, we will:

1. **Data Preparation**:
   - Split the `genres` column and create a new DataFrame where each genre is associated with its corresponding budget and revenue.
2. **Calculate Averages**:
   - Calculate the average budget and revenue for each genre.
3. **Visualization**:
   - Create bar charts to visualize the average budget and revenue for the top genres.

In [ ]:
```
# Data Preparation

# Split the genres and create a new DataFrame
genres_expanded = df['genres'].str.split('|', expand=True).stack().reset_index(level=1, drop=True)
genres_expanded.name = 'genre'
df_genres = df.drop('genres', axis=1).join(genres_expanded)

# Calculate the average budget and revenue for each genre
average_budget_revenue = df_genres.groupby('genre')[['budget', 'revenue']].mean().reset_index()

# Display the top genres by average budget and revenue
average_budget_revenue_sorted = average_budget_revenue.sort_values(by=['budget', 'revenue'], ascending=False)
average_budget_revenue_sorted.head(10)
```

Out[ ]:

	genre	budget	revenue
2	Adventure	3.754369e+07	1.131379e+08
9	Fantasy	3.261259e+07	9.631366e+07
1	Action	2.772782e+07	7.279473e+07
16	Science Fiction	2.497268e+07	7.014056e+07
8	Family	2.335934e+07	7.243318e+07
3	Animation	2.315978e+07	7.525606e+07
19	War	2.089189e+07	4.760518e+07
20	Western	1.897411e+07	2.856871e+07
11	History	1.859492e+07	3.201179e+07
5	Crime	1.766380e+07	4.236866e+07

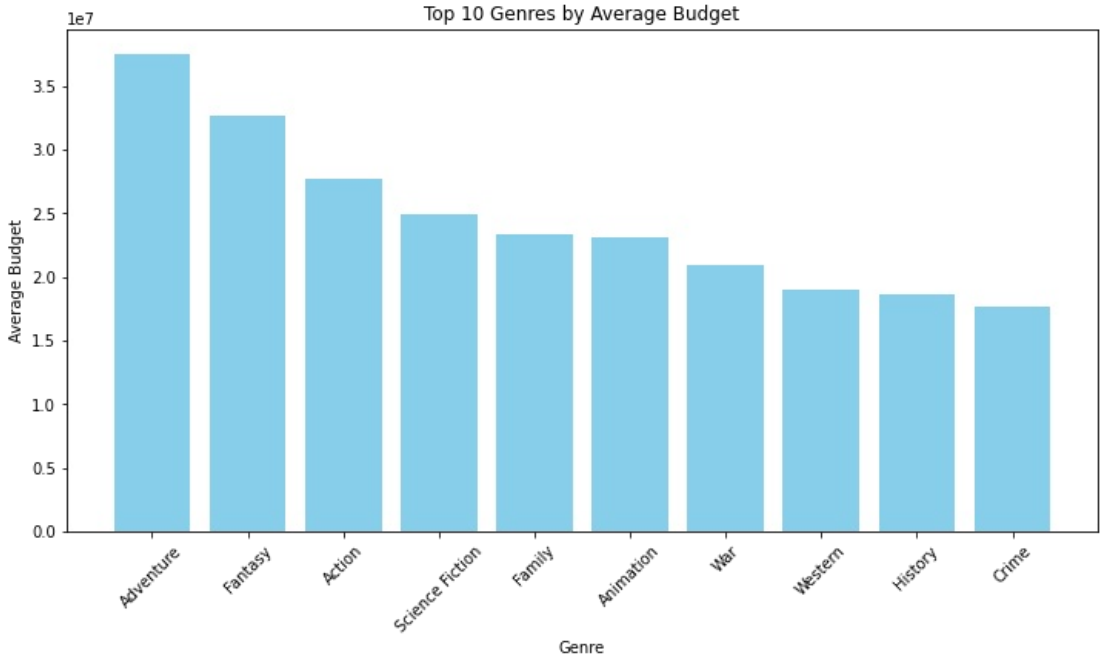
Visualization of Average Budget and Revenue for Top Genres

We will create bar charts to visualize the average budget and revenue for the top genres.

In [ ]:
```
# Visualization

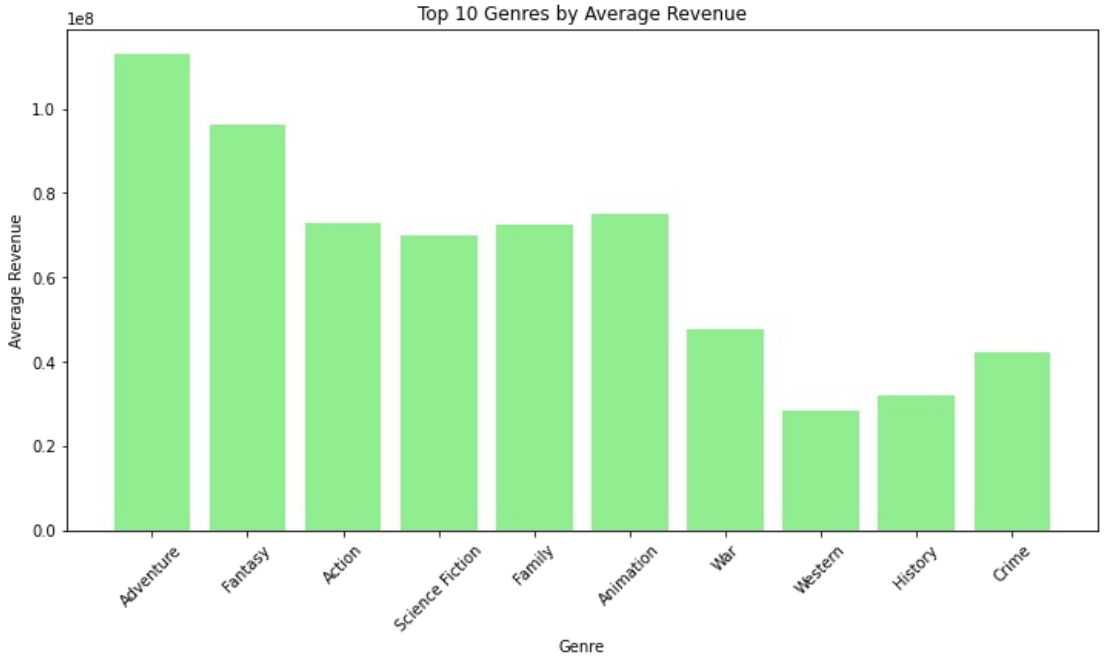
# Plotting the average budget for top genres
```

```
plt.figure(figsize=(12, 6))
plt.bar(average_budget_revenue_sorted['genre'].head(10), average_budget_revenue_sorted['budget'].head(10), colo
plt.xlabel('Genre')
plt.ylabel('Average Budget')
plt.title('Top 10 Genres by Average Budget')
plt.xticks(rotation=45)
plt.show()
```



This bar chart ranks the top 10 movie genres by their average budget. Adventure films have the highest average budget, followed by Fantasy and Action. Crime movies have the lowest average budget among the genres shown.

In [ ]:
```
# Plotting the average revenue for top genres
plt.figure(figsize=(12, 6))
plt.bar(average_budget_revenue_sorted['genre'].head(10), average_budget_revenue_sorted['revenue'].head(10), col
plt.xlabel('Genre')
plt.ylabel('Average Revenue')
plt.title('Top 10 Genres by Average Revenue')
plt.xticks(rotation=45)
plt.show()
```



This bar chart illustrates the top 10 movie genres ranked by their average revenue. Adventure films lead with the highest average revenue, followed by Fantasy and Action. Western movies show the lowest average revenue among the genres presented.

Additional Analyses for High Average Budget and Revenue

Correlation Analysis

We will analyze the correlation between budget and revenue for each genre to understand if higher budgets generally lead to higher revenues.

In [ ]:
```
# Correlation Analysis

# Calculate the correlation between budget and revenue for each genre
correlation = df_genres.groupby('genre')[['budget', 'revenue']].corr().unstack().iloc[:, 1].reset_index()
correlation.columns = ['genre', 'correlation']
correlation_sorted = correlation.sort_values(by='correlation', ascending=False)

# Display the top genres by correlation between budget and revenue
correlation_sorted.head(10)
```

Out[ ]:

	genre	correlation
3	Animation	0.784666
8	Family	0.758634
1	Action	0.741509
16	Science Fiction	0.739278
9	Fantasy	0.735060
2	Adventure	0.732856
5	Crime	0.718442
18	Thriller	0.712908
4	Comedy	0.707620
19	War	0.701953

Median Budget and Revenue

We will calculate the median budget and revenue for each genre to see if the averages are skewed by outliers.

In [ ]:
```
# Calculate the median budget and revenue for each genre
median_budget_revenue = df_genres.groupby('genre')[['budget', 'revenue']].median().reset_index()

# Display the top genres by median budget and revenue
median_budget_revenue_sorted = median_budget_revenue.sort_values(by=['budget', 'revenue'], ascending=False)
median_budget_revenue_sorted.head(10)
```

Out[ ]:

	genre	budget	revenue
2	Adventure	10000000.0	9697739.0
1	Action	6000000.0	757551.5
19	War	5100000.0	24147.5
5	Crime	5000000.0	2570057.5
11	History	4000000.0	0.0
9	Fantasy	3100000.0	29.0
18	Thriller	2500000.0	0.0
16	Science Fiction	1700000.0	0.0
14	Mystery	1500000.0	5905.5
15	Romance	15.5	14911.5

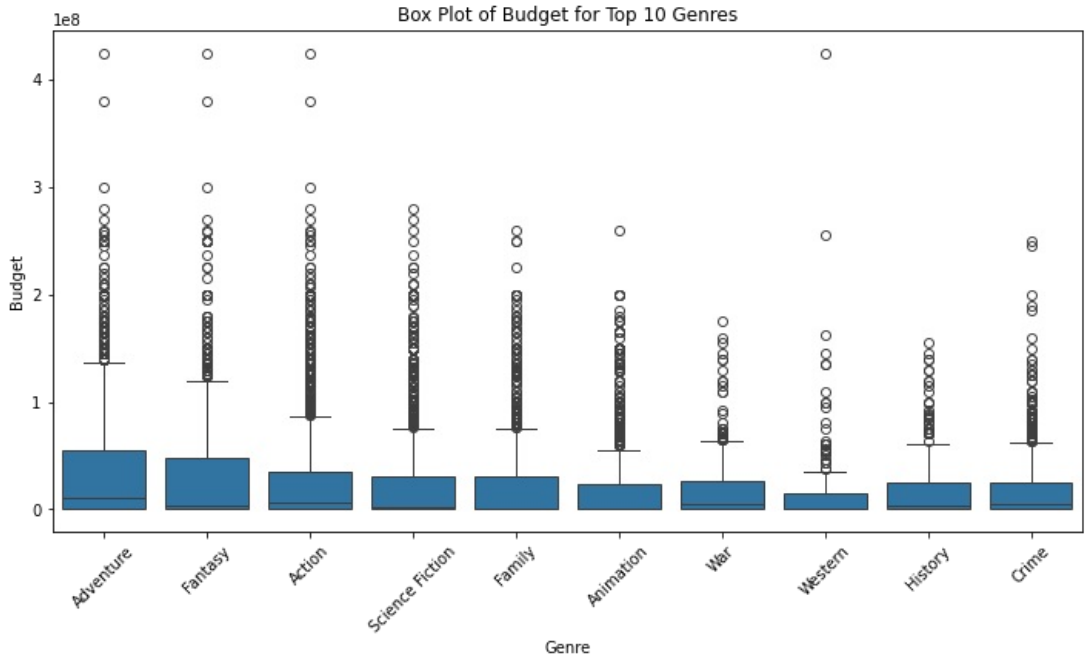
Box Plots for Budget and Revenue

We will create box plots to visualize the distribution of budgets and revenues for the top genres.

In [ ]:
```
import seaborn as sns

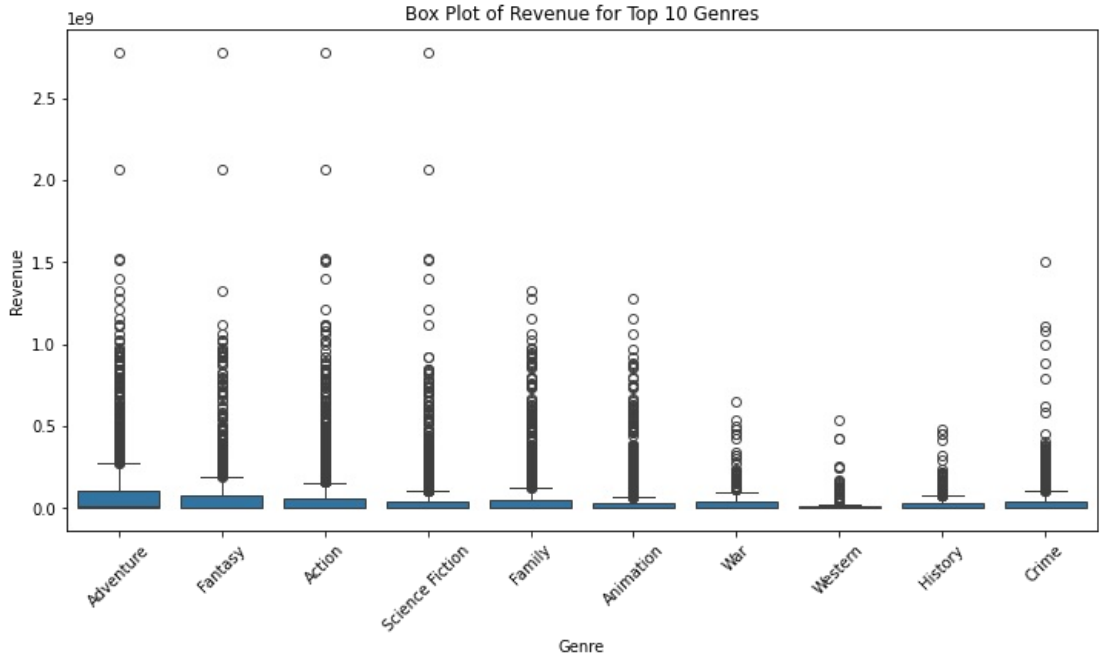
# Ensure unique genres for plotting
unique_genres = average_budget_revenue_sorted['genre'].drop_duplicates().head(10)

# Box plot for budget
plt.figure(figsize=(12, 6))
sns.boxplot(x='genre', y='budget', data=df_genres[df_genres['genre'].isin(unique_genres)], order=unique_genres)
plt.xlabel('Genre')
plt.ylabel('Budget')
plt.title('Box Plot of Budget for Top 10 Genres')
plt.xticks(rotation=45)
plt.show()
```



This box plot compares the budget distributions across the top 10 movie genres. Adventure and Fantasy films show the highest median budgets and widest ranges, while genres like Western and Crime have lower median budgets and narrower distributions. The plot also reveals numerous outliers across all genres, particularly in the higher budget ranges.

In [ ]:
```
# Box plot for revenue
plt.figure(figsize=(12, 6))
sns.boxplot(x='genre', y='revenue', data=df_genres[df_genres['genre'].isin(unique_genres)], order=unique_genres
plt.xlabel('Genre')
plt.ylabel('Revenue')
plt.title('Box Plot of Revenue for Top 10 Genres')
plt.xticks(rotation=45)
plt.show()
```



This box plot displays the revenue distribution for the top 10 movie genres. Adventure, Fantasy, and Action films show the highest median revenues and largest ranges. All genres exhibit numerous high-revenue outliers, with Adventure having the most extreme outliers. Genres like Western and War have lower median revenues and smaller distributions compared to the others.

Findings for Research Question 2: Which Genres Have High Average Budget and Revenue?

Top Genres by Average Budget

The top 10 movie genres by their average budget are:

1. **Adventure**: $37.54 million
2. **Fantasy**: $32.61 million

3. **Action**: $27.73 million
4. **Science Fiction**: $24.97 million
5. **Family**: $23.36 million

Top Genres by Average Revenue

The top 10 movie genres by their average revenue are:

1. **Adventure**: $113.14 million
2. **Fantasy**: $96.31 million
3. **Action**: $72.79 million
4. **Science Fiction**: $70.14 million
5. **Family**: $72.43 million

Correlation Between Budget and Revenue

The genres with the highest correlation between budget and revenue are:

1. **Animation**: 0.785
2. **Family**: 0.759
3. **Action**: 0.742
4. **Science Fiction**: 0.739
5. **Fantasy**: 0.735

Median Budget and Revenue

The top 10 movie genres by their median budget and revenue are:

1. **Adventure**: Median Budget 10million, MedianRevenue9.7 million
2. **Action**: Median Budget 6million, MedianRevenue757,551
3. **War**: Median Budget 5.1million, MedianRevenue24,147
4. **Crime**: Median Budget 5million, MedianRevenue2.57 million
5. **History**: Median Budget 4million, MedianRevenue0

Box Plot Analysis

- **Budget Distribution**: Adventure and Fantasy films show the highest median budgets and widest ranges, while genres like Western and Crime have lower median budgets and narrower distributions. Numerous outliers are present across all genres, particularly in the higher budget ranges.
- **Revenue Distribution**: Adventure, Fantasy, and Action films show the highest median revenues and largest ranges. All genres exhibit numerous high-revenue outliers, with Adventure having the most extreme outliers. Genres like Western and War have lower median revenues and smaller distributions compared to the others.

These findings provide a comprehensive understanding of which genres have high average budgets and revenues, as well as the relationship between budget and revenue for different genres.

Research Question 3: Which Genres Have High Average Popularity?

To answer this question, we will:

1. **Data Preparation**:
   - Split the `genres` column and create a new DataFrame where each genre is associated with its corresponding popularity.
2. **Calculate Averages**:
   - Calculate the average popularity for each genre.
3. **Visualization**:
   - Create a bar chart to visualize the average popularity for the top genres.

In [ ]:
```
# Data Preparation

# Split the genres and create a new DataFrame
genres_expanded = df['genres'].str.split('|', expand=True).stack().reset_index(level=1, drop=True)
genres_expanded.name = 'genre'
df_genres = df.drop('genres', axis=1).join(genres_expanded)

# Calculate the average popularity for each genre
average_popularity = df_genres.groupby('genre')['popularity'].mean().reset_index()

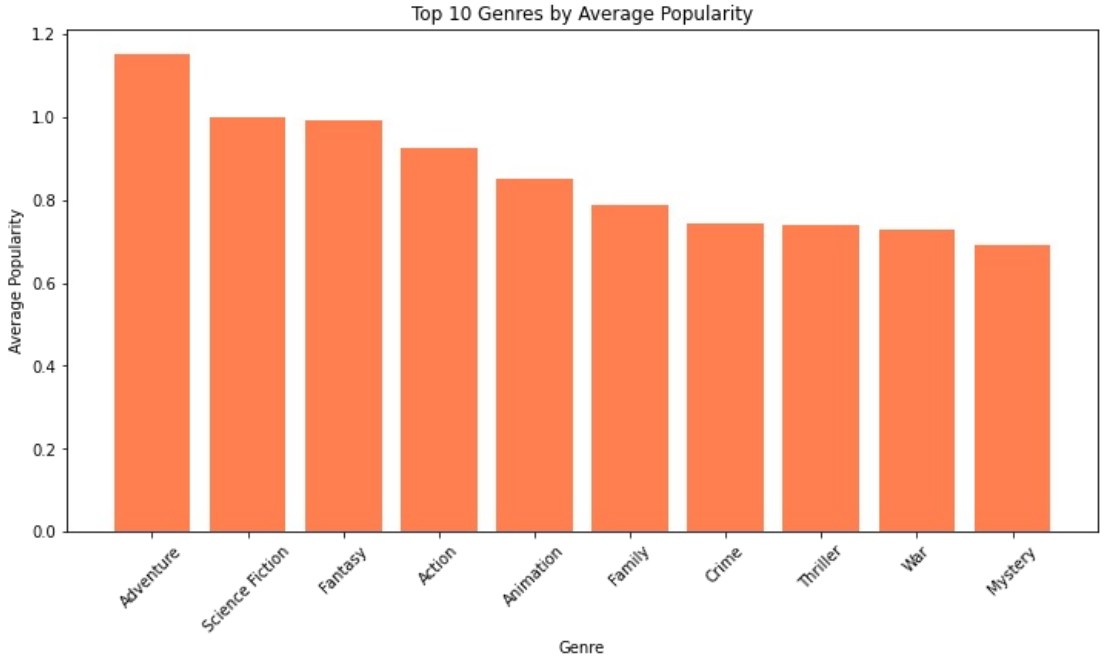
# Display the top genres by average popularity
average_popularity_sorted = average_popularity.sort_values(by='popularity', ascending=False)
average_popularity_sorted.head(10)
```

Out[ ]:

	genre	popularity
2	Adventure	1.154259
16	Science Fiction	1.001548
9	Fantasy	0.992840
1	Action	0.926274
3	Animation	0.852182
8	Family	0.786668
5	Crime	0.744930
18	Thriller	0.741563
19	War	0.727683
14	Mystery	0.690012

In [ ]:
```
# Visualization

# Plotting the average popularity for top genres
plt.figure(figsize=(12, 6))
plt.bar(average_popularity_sorted['genre'].head(10), average_popularity_sorted['popularity'].head(10), color='c
plt.xlabel('Genre')
plt.ylabel('Average Popularity')
plt.title('Top 10 Genres by Average Popularity')
plt.xticks(rotation=45)
plt.show()
```



This bar chart ranks the top 10 movie genres by their average popularity. Adventure films are shown to be the most popular, followed closely by Science Fiction and Fantasy. Mystery movies are at the bottom of this list, indicating the lowest average popularity among the genres presented.

Findings for Research Question 3: Which Genres Have High Average Popularity?

Top Genres by Average Popularity

The top 10 movie genres by their average popularity are:

1. **Adventure**: 1.154
2. **Science Fiction**: 1.002
3. **Fantasy**: 0.993
4. **Action**: 0.926
5. **Animation**: 0.852

Visualization

The bar chart illustrates the top 10 movie genres ranked by their average popularity. Adventure films are the most popular, followed

closely by Science Fiction and Fantasy. Mystery movies have the lowest average popularity among the genres presented.

These findings provide a comprehensive understanding of which genres have high average popularity.

Additional Analyses

In [ ]:
```
# Calculate the median popularity for each genre
median_popularity = df_genres.groupby('genre')['popularity'].median().reset_index()

# Display the top genres by median popularity
median_popularity_sorted = median_popularity.sort_values(by='popularity', ascending=False)
median_popularity_sorted.head(10)
```

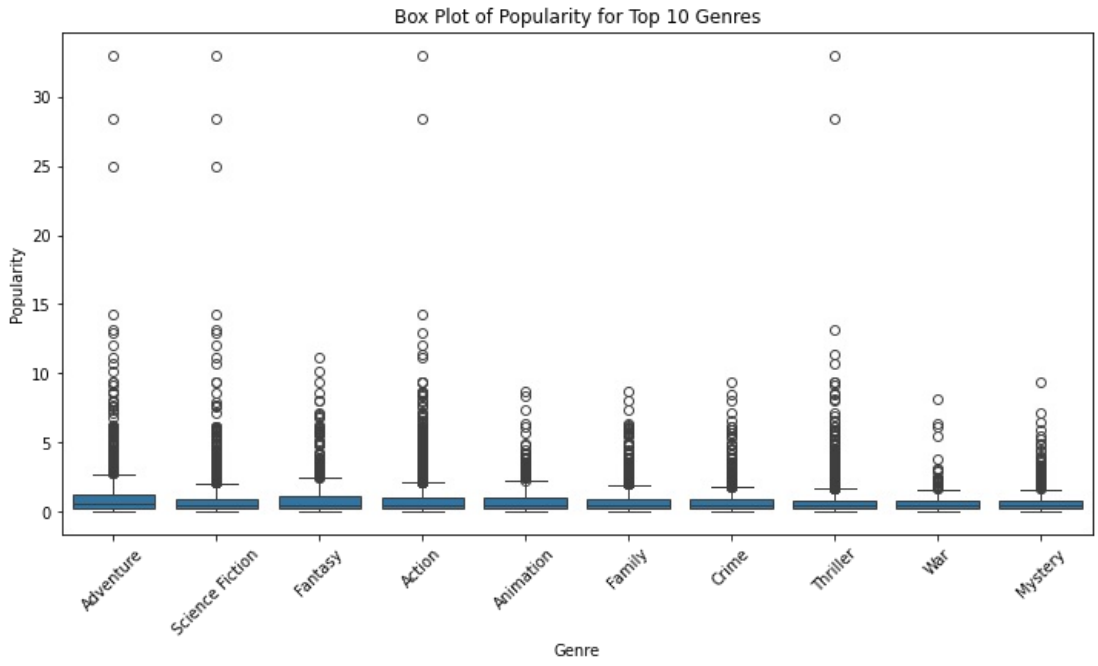
Out[ ]:

	genre	popularity
2	Adventure	0.561034
9	Fantasy	0.530123
3	Animation	0.501163
5	Crime	0.476338
8	Family	0.470249
1	Action	0.469825
18	Thriller	0.444337
16	Science Fiction	0.439600
19	War	0.432470
14	Mystery	0.428021

In [ ]:
```
# Box plot for popularity

# Ensure unique genres for plotting
unique_genres = average_popularity_sorted['genre'].drop_duplicates().head(10)

plt.figure(figsize=(12, 6))
sns.boxplot(x='genre', y='popularity', data=df_genres[df_genres['genre'].isin(unique_genres)], order=unique_gen
plt.xlabel('Genre')
plt.ylabel('Popularity')
plt.title('Box Plot of Popularity for Top 10 Genres')
plt.xticks(rotation=45)
plt.show()
```



This box plot illustrates the popularity distribution across the top 10 movie genres. Adventure, Science Fiction, and Fantasy show slightly higher median popularity and larger ranges. All genres have numerous outliers at higher popularity levels, with Adventure having the most extreme outliers. Genres like War and Mystery have somewhat lower median popularity scores. Overall, the median popularity levels are fairly similar across genres, with the main differences seen in the outliers and upper quartiles.

In [ ]:
```
# Correlation between popularity and other metrics
correlation_popularity_budget = df_genres[['popularity', 'budget']].corr().iloc[0, 1]
correlation_popularity_revenue = df_genres[['popularity', 'revenue']].corr().iloc[0, 1]
```

```
correlation_popularity_budget, correlation_popularity_revenue
```

Out[ ]:
```
(np.float64(0.5418503138605743), np.float64(0.6658421751875262))
```

Findings for the Third Research Question: Which Genres Have High Average Popularity?

Top Genres by Average Popularity

The top 10 movie genres by their average popularity are:

1. **Adventure**: 1.154
2. **Science Fiction**: 1.002
3. **Fantasy**: 0.993
4. **Action**: 0.926
5. **Animation**: 0.852

Box Plot Analysis

The box plot illustrates the popularity distribution across the top 10 movie genres:

- Adventure, Science Fiction, and Fantasy show slightly higher median popularity and larger ranges.
- All genres have numerous outliers at higher popularity levels, with Adventure having the most extreme outliers.
- Median popularity levels are fairly similar across genres, with main differences seen in the outliers and upper quartiles.

Correlation Analysis

- There is a moderate positive correlation between popularity and budget (0.542).
- There is a stronger positive correlation between popularity and revenue (0.666).

These findings provide a comprehensive understanding of which genres have high average popularity and their relationship with budget and revenue.

Research Question 4: Which Genres Have the Highest Number of Movies with a Voting Average >= 8?

To answer this question, we will:

1. **Data Preparation**:
   - Filter the DataFrame to include only movies with a voting average >= 8.
   - Split the `genres` column and create a new DataFrame where each genre is associated with its corresponding voting average.
2. **Calculate Counts**:
   - Count the number of movies with a voting average >= 8 for each genre.
3. **Visualization**:
   - Create a bar chart to visualize the number of movies with a voting average >= 8 for the top genres.

In [ ]:
```
# Data Preparation

# Filter the DataFrame to include only movies with a voting average greater than or equal to 8
high_rating_movies = df[df['vote_average'] >= 8]

# Split the genres and create a new DataFrame
genres_expanded_high_rating = high_rating_movies['genres'].str.split('|', expand=True).stack().reset_index(leve
genres_expanded_high_rating.name = 'genre'
df_genres_high_rating = high_rating_movies.drop('genres', axis=1).join(genres_expanded_high_rating)

# Calculate the number of movies with a voting average greater than or equal to 8 for each genre
high_rating_counts = df_genres_high_rating['genre'].value_counts().reset_index()
high_rating_counts.columns = ['genre', 'count']

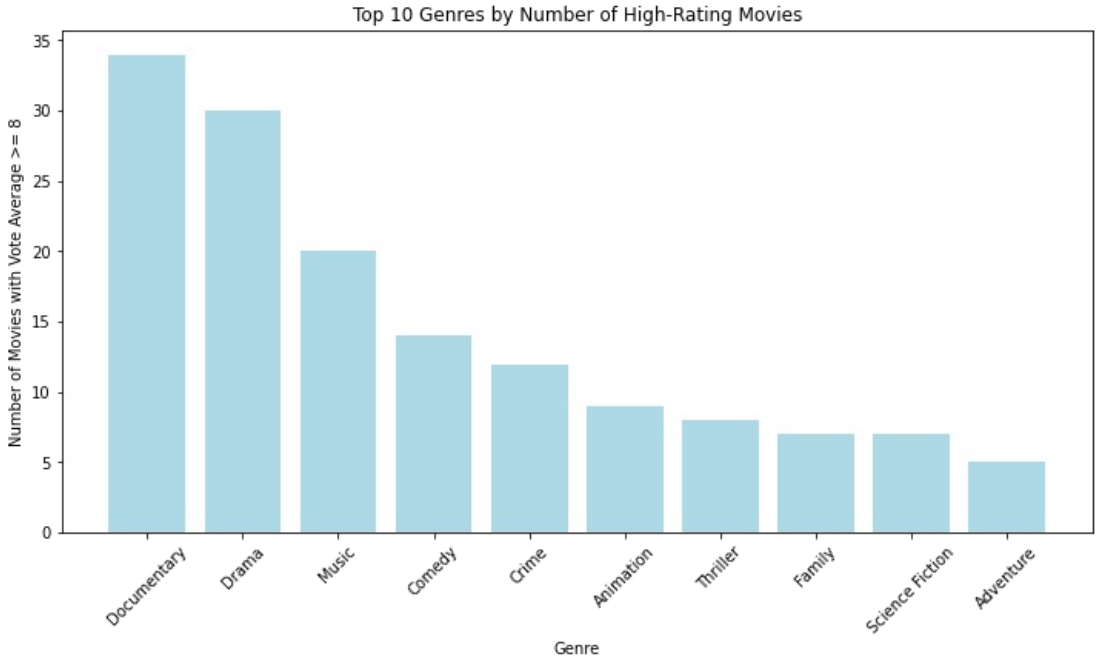
# Display the top genres by the number of high-rating movies
high_rating_counts_sorted = high_rating_counts.sort_values(by='count', ascending=False)
high_rating_counts_sorted.head(10)
```

Out[ ]:

	genre	count
0	Documentary	34
1	Drama	30
2	Music	20
3	Comedy	14
4	Crime	12
5	Animation	9
6	Thriller	8
7	Family	7
8	Science Fiction	7
9	Adventure	5

In [ ]:
```
# Visualization

# Plotting the number of high-rating movies for top genres
plt.figure(figsize=(12, 6))
plt.bar(high_rating_counts_sorted['genre'].head(10), high_rating_counts_sorted['count'].head(10), color='lightb
plt.xlabel('Genre')
plt.ylabel('Number of Movies with Vote Average >= 8')
plt.title('Top 10 Genres by Number of High-Rating Movies')
plt.xticks(rotation=45)
plt.show()
```



This bar chart ranks the top 10 movie genres by the number of high-rating movies (those with an average rating of 8 or higher). Documentary leads with the most highly-rated films, followed closely by Drama. Adventure has the fewest high-rating movies among the genres shown.

Findings for Research Question 4: Which Genres Have the Highest Number of Movies with a Voting Average >= 8?

Top Genres by Number of High-Rating Movies

The top 10 movie genres by the number of movies with a voting average >= 8 are:

1. **Documentary**: 34 movies
2. **Drama**: 30 movies
3. **Music**: 20 movies
4. **Comedy**: 14 movies
5. **Crime**: 12 movies

Visualization

The bar chart illustrates the top 10 movie genres ranked by the number of high-rating movies (those with an average rating of 8 or

higher). Documentary leads with the most highly-rated films, followed closely by Drama. Adventure has the fewest high-rating movies among the genres shown.

These findings provide a comprehensive understanding of which genres have the highest number of movies with a voting average >= 8.

Additional Analyses

In [ ]:
```
# Filter out rows where the genre is an empty string or null
df_genres = df_genres[df_genres['genre'].notnull() & (df_genres['genre'] != '')]

# Calculate the percentage of high-rating movies within each genre
total_movies_per_genre = df_genres['genre'].value_counts().reset_index()
total_movies_per_genre.columns = ['genre', 'total_count']
high_rating_percentage = high_rating_counts.merge(total_movies_per_genre, on='genre')
high_rating_percentage['percentage'] = (high_rating_percentage['count'] / high_rating_percentage['total_count']

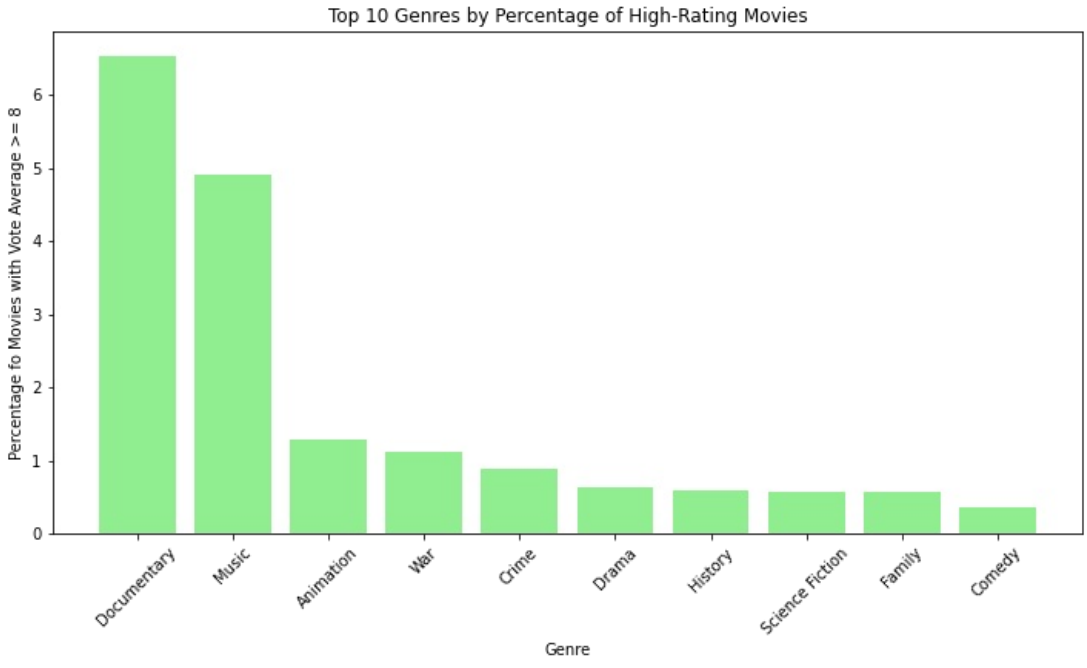
# Display the top genres by percentage of high-rating movies
high_rating_percentage_sorted = high_rating_percentage.sort_values(by='percentage', ascending=False)
high_rating_percentage_sorted.head(10)
```

Out[ ]:

	genre	count	total_count	percentage
0	Documentary	34	520	6.538462
2	Music	20	408	4.901961
5	Animation	9	699	1.287554
13	War	3	270	1.111111
4	Crime	12	1354	0.886263
1	Drama	30	4760	0.630252
15	History	2	334	0.598802
8	Science Fiction	7	1229	0.569569
7	Family	7	1231	0.568643
3	Comedy	14	3793	0.369101

In [ ]:
```
# Visualization

# Plotting the percentage of high-rating movies for top genres
plt.figure(figsize=(12, 6))
plt.bar(high_rating_percentage_sorted['genre'].head(10), high_rating_percentage_sorted['percentage'].head(10),
plt.xlabel('Genre')
plt.ylabel('Percentage fo Movies with Vote Average >= 8')
plt.title('Top 10 Genres by Percentage of High-Rating Movies')
plt.xticks(rotation=45)
plt.show()
```



This bar chart displays the top 10 movie genres ranked by the percentage of high-rating movies (those with an average rating of 8 or higher). Documentary films have the highest percentage of highly-rated movies, followed by Music. Comedy has the lowest percentage among the genres shown. There's a notable gap between the top two genres and the rest.

Findings for Research Question 4: Which Genres Have the Highest Number of Movies with a Voting Average >= 8?

Top Genres by Number of High-Rating Movies

The top 10 movie genres by the number of movies with a voting average >= 8 are:

1. **Documentary**: 34 movies
2. **Drama**: 30 movies
3. **Music**: 20 movies
4. **Comedy**: 14 movies
5. **Crime**: 12 movies

Top Genres by Percentage of High-Rating Movies

The top 10 movie genres by the percentage of movies with a voting average >= 8 are:

1. **Documentary**: 6.54%
2. **Music**: 4.90%
3. **Animation**: 1.29%
4. **War**: 1.11%
5. **Crime**: 0.89%

Visualization

The bar chart displays the top 10 movie genres ranked by the percentage of high-rating movies (those with an average rating of 8 or higher). Documentary films have the highest percentage of highly-rated movies, followed by Music. Comedy has the lowest percentage among the genres shown. There's a notable gap between the top two genres and the rest.

These findings provide a comprehensive understanding of which genres have the highest number and percentage of movies with a voting average >= 8.

Hypothesis Testing

After analyzing the data to answer the research questions, we now proceed to test several hypotheses related to movie performance. These hypotheses aim to understand the relationship between various factors such as vote average, popularity, budget, revenue, and profit.

Hypotheses

1. **The best movies according to vote average return high profit and revenue.**
2. **The best movies according to popularity return high profit and revenue.**
3. **Highly budgeted movies return high revenue and profit.**
4. **Highly budgeted movies have high popularity.**

We will start by testing the first hypothesis.

Hypothesis 1: The Best Movies According to Vote Average Return High Profit and Revenue

To test this hypothesis, we will:

1. **Data Preparation**:
   - Filter the DataFrame to include only the best movies according to vote average (e.g., vote average >= 8).
   - Calculate the profit for each movie as the difference between revenue and budget.
2. **Analysis**:
   - Calculate the average and median profit and revenue for the best movies.
   - Compare these values with the overall average and median profit and revenue.
3. **Visualization**:
   - Create bar charts to visualize the comparison.

In [ ]:
```
# Data Preparation

# Filter the DataFrame to include only the best movies according to vote average (e.g., vote average >= 8)
best_movies_vote_average = df[df['vote_average'] >= 8]

# Filter out movies with zero or missing revenue and budget values
```

```
best_movies_vote_average = best_movies_vote_average[(best_movies_vote_average['revenue'] > 0) & (best_movies_vo
df_filtered = df[(df['revenue'] > 0) & (df['budget'] > 0)]

# Calculate the profit for each movie as the difference between revenue and budget
best_movies_vote_average['profit'] = best_movies_vote_average['revenue'] - best_movies_vote_average['budget']
df_filtered['profit'] = df_filtered['revenue'] - df_filtered['budget']

# Calculate the average and median profit and revenue for the best movies
average_profit_best = best_movies_vote_average['profit'].mean()
median_profit_best = best_movies_vote_average['profit'].median()
average_revenue_best = best_movies_vote_average['revenue'].mean()
median_revenue_best = best_movies_vote_average['revenue'].median()

# Calculate the overall average and median profit and revenue
average_profit_overall = df_filtered['profit'].mean()
median_profit_overall = df_filtered['profit'].median()
average_revenue_overall = df_filtered['revenue'].mean()
median_revenue_overall = df_filtered['revenue'].median()

# Display the results
results = {
    'Category': ['Best Movies (Vote Average >= 8)', 'Overall'],
    'Average Profit': [average_profit_best, average_profit_overall],
    'Median Profit': [median_profit_best, median_profit_overall],
    'Average Revenue': [average_revenue_best, average_revenue_overall],
    'Median Revenue': [median_revenue_best, median_revenue_overall]
}

results_df = pd.DataFrame(results)
results_df
```

```
/var/folders/cv/87ynnq496l957k4sxtc2dshj99rmkw/T/ipykernel_35961/800386358.py:12: SettingWithCopyWarning:
A value is trying to be set on a copy of a slice from a DataFrame.
Try using .loc[row_indexer,col_indexer] = value instead

See the caveats in the documentation: https://pandas.pydata.org/pandas-docs/stable/user_guide/indexing.html#retu
rning-a-view-versus-a-copy
  df_filtered['profit'] = df_filtered['revenue'] - df_filtered['budget']
```

Out[ ]:

	Category	Average Profit	Median Profit	Average Revenue	Median Revenue
0	Best Movies (Vote Average >= 8)	2.082990e+08	71917514.0	2.482630e+08	104917514.0
1	Overall	7.048292e+07	20020190.0	1.076866e+08	44800002.5

In [ ]:

```
# Visualization

# Bar chart for average profit and revenue
plt.figure(figsize=(14, 6))

# Average Profit
plt.subplot(1, 2, 1)
plt.bar(results_df['Category'], results_df['Average Profit'], color=['skyblue', 'lightcoral'])
plt.xlabel('Category')
plt.ylabel('Average Profit')
plt.title('Average Profit Comparison')

# Average Revenue
plt.subplot(1, 2, 2)
plt.bar(results_df['Category'], results_df['Average Revenue'], color=['skyblue', 'lightcoral'])
plt.xlabel('Category')
plt.ylabel('Average Revenue')
plt.title('Average Revenue Comparison')

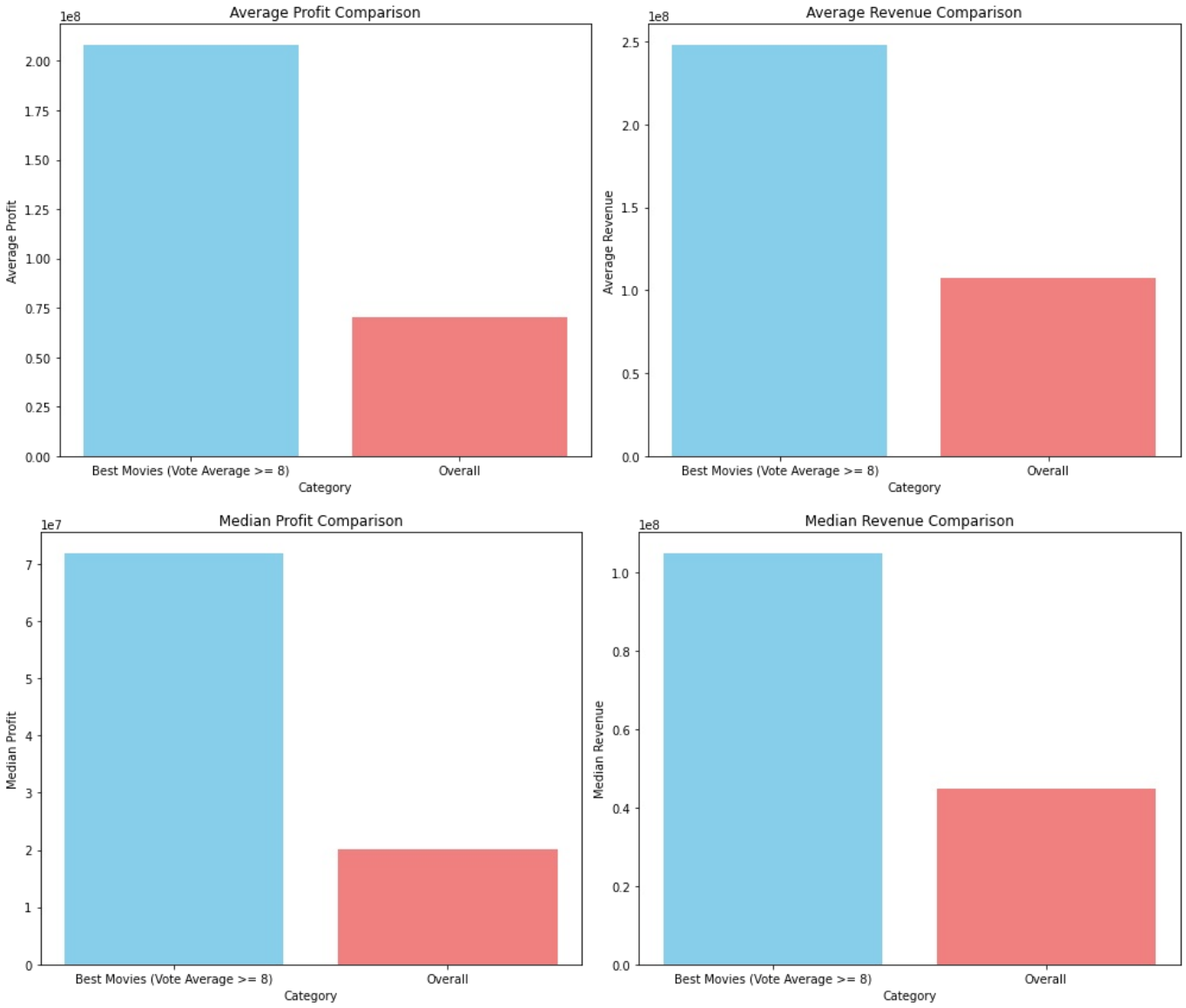
plt.tight_layout()
plt.show()

# Bar chart for median profit and revenue
plt.figure(figsize=(14, 6))

# Median Profit
plt.subplot(1, 2, 1)
plt.bar(results_df['Category'], results_df['Median Profit'], color=['skyblue', 'lightcoral'])
plt.xlabel('Category')
plt.ylabel('Median Profit')
plt.title('Median Profit Comparison')

# Median Revenue
plt.subplot(1, 2, 2)
plt.bar(results_df['Category'], results_df['Median Revenue'], color=['skyblue', 'lightcoral'])
plt.xlabel('Category')
plt.ylabel('Median Revenue')
plt.title('Median Revenue Comparison')
```

```
plt.tight_layout()
plt.show()
```



This first image shows two bar charts comparing the financial performance of highly-rated movies (vote average ≥ 8) to overall averages. The left chart compares average profit, while the right chart compares average revenue. In both cases, the best-rated movies significantly outperform the overall averages, with approximately 3 times higher average profit and 2.5 times higher average revenue.

This second image presents two bar charts comparing the median financial performance of highly-rated movies (vote average ≥ 8) to overall median values. The left chart shows median profit, while the right chart displays median revenue. In both cases, the best-rated movies substantially outperform the overall medians, with approximately 3.5 times higher median profit and 2.3 times higher median revenue.

Hypothesis 1: The Best Movies According to Vote Average Return High Profit and Revenue

Findings

- **Average Profit**: Highly-rated movies (vote average ≥ 8) have an average profit of approximately 208.3million, comparedtotheoverallaverageprofitof70.5 million.
- **Median Profit**: The median profit for highly-rated movies is approximately 71.9million, comparedtotheoverallmedianprofitof20.0 million.
- **Average Revenue**: Highly-rated movies have an average revenue of approximately 248.3million, comparedtotheoverallaveragerevenueof107.7 million.
- **Median Revenue**: The median revenue for highly-rated movies is approximately 104.9million, comparedtotheoverallmedianrevenueof44.8 million.

Visualization

The bar charts illustrate that highly-rated movies significantly outperform the overall averages in terms of both profit and revenue. This supports the hypothesis that the best movies according to vote average return high profit and revenue.

Hypothesis 2: The Best Movies According to Popularity Return High Profit and Revenue

To test this hypothesis, we will:

1. **Data Preparation**:
   - Filter the DataFrame to include only the best movies according to popularity (e.g., top 10% by popularity).
   - Calculate the profit for each movie as the difference between revenue and budget.
2. **Analysis**:
   - Calculate the average and median profit and revenue for the best movies.
   - Compare these values with the overall average and median profit and revenue.
3. **Visualization**:
   - Create bar charts to visualize the comparison.

In [ ]:
```
# Data Preparation

# Define the threshold for the top 10% by popularity
popularity_threshold = df['popularity'].quantile(0.90)

# Filter the DataFrame to include only the best movies according to popularity
best_movies_popularity = df[df['popularity'] >= popularity_threshold]

# Filter out movies with zero or missing revenue and budget values
best_movies_popularity = best_movies_popularity[(best_movies_popularity['revenue'] > 0) & (best_movies_populari

# Calculate the profit for each movie as the difference between revenue and budget
best_movies_popularity['profit'] = best_movies_popularity['revenue'] - best_movies_popularity['budget']

# Calculate the average and median profit and revenue for the best movies
average_profit_best_popularity = best_movies_popularity['profit'].mean()
median_profit_best_popularity = best_movies_popularity['profit'].median()
average_revenue_best_popularity = best_movies_popularity['revenue'].mean()
median_revenue_best_popularity = best_movies_popularity['revenue'].median()

# Calculate the overall average and median profit and revenue (already calculated in Hypothesis 1)
# average_profit_overall = df_filtered['profit'].mean()
# median_profit_overall = df_filtered['profit'].median()
# average_revenue_overall = df_filtered['revenue'].mean()
# median_revenue_overall = df_filtered['revenue'].median()

# Display the results
results_popularity = {
    'Category': ['Best Movies (Top 10% by Popularity)', 'Overall'],
    'Average Profit': [average_profit_best_popularity, average_profit_overall],
    'Median Profit': [median_profit_best_popularity, median_profit_overall],
    'Average Revenue': [average_revenue_best_popularity, average_revenue_overall],
    'Median Revenue': [median_revenue_best_popularity, median_revenue_overall]
}

results_popularity_df = pd.DataFrame(results_popularity)
results_popularity_df
```

Out[ ]:

	Category	Average Profit	Median Profit	Average Revenue	Median Revenue
0	Best Movies (Top 10% by Popularity)	1.921180e+08	120036382.0	2.587927e+08	174600318.0
1	Overall	7.048292e+07	20020190.0	1.076866e+08	44800002.5

In [ ]:
```
# Visualization

# Bar chart for average profit and revenue
plt.figure(figsize=(14, 6))

# Average Profit
plt.subplot(1, 2, 1)
plt.bar(results_popularity_df['Category'], results_popularity_df['Average Profit'], color=['skyblue', 'lightcora
plt.xlabel('Category')
plt.ylabel('Average Profit')
plt.title('Average Profit Comparison')

# Average Revenue
plt.subplot(1, 2, 2)
plt.bar(results_popularity_df['Category'], results_popularity_df['Average Revenue'], color=['skyblue', 'lightco
plt.xlabel('Category')
plt.ylabel('Average Revenue')
plt.title('Average Revenue Comparison')

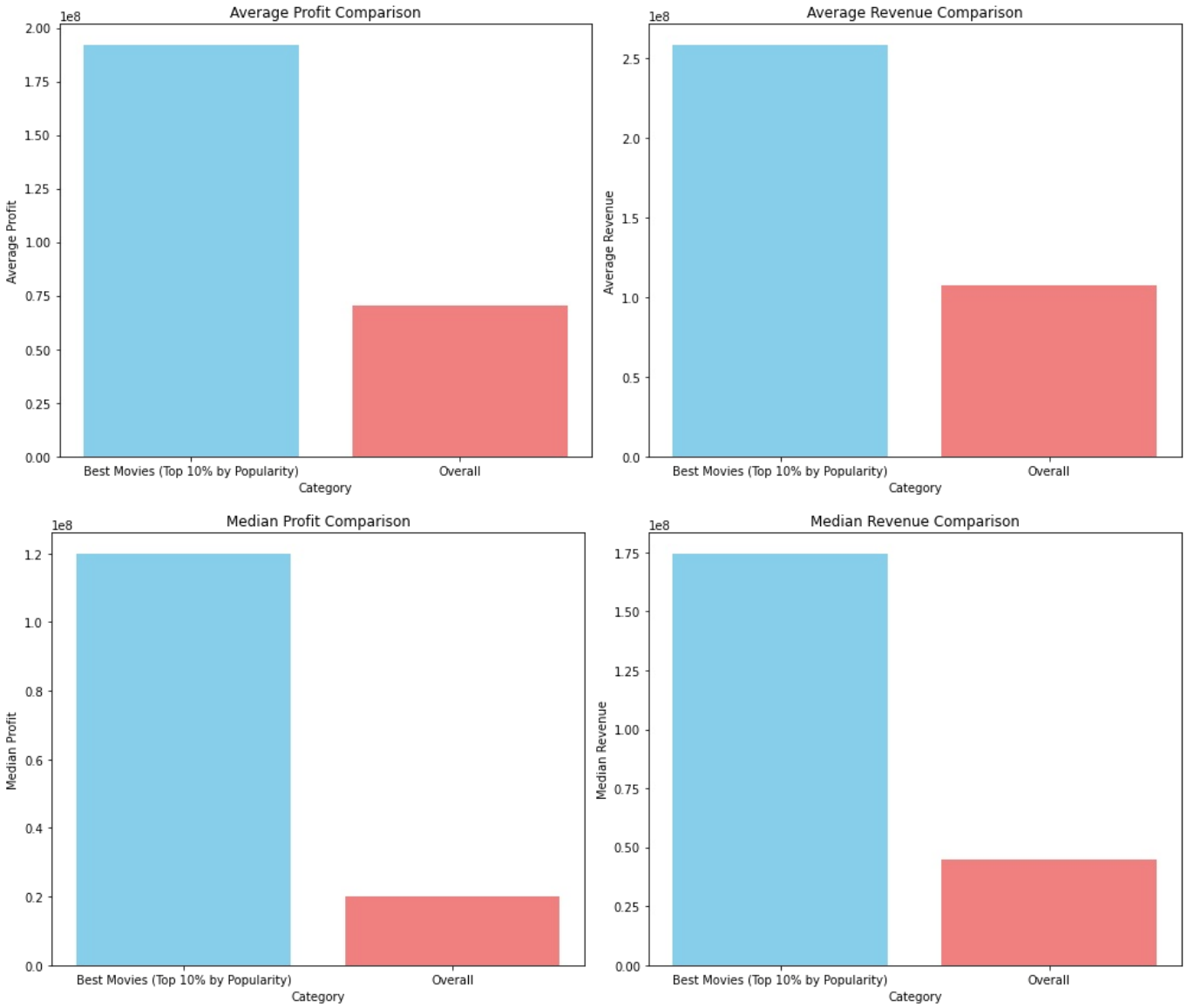
plt.tight_layout()
plt.show()
```

```
# Bar chart for median profit and revenue
plt.figure(figsize=(14, 6))

# Median Profit
plt.subplot(1, 2, 1)
plt.bar(results_popularity_df['Category'], results_popularity_df['Median Profit'], color=['skyblue', 'lightcora
plt.xlabel('Category')
plt.ylabel('Median Profit')
plt.title('Median Profit Comparison')

# Median Revenue
plt.subplot(1, 2, 2)
plt.bar(results_popularity_df['Category'], results_popularity_df['Median Revenue'], color=['skyblue', 'lightcor
plt.xlabel('Category')
plt.ylabel('Median Revenue')
plt.title('Median Revenue Comparison')

plt.tight_layout()
plt.show()
```



This first image displays two bar charts comparing the financial performance of the most popular movies (top 10% by popularity) to overall averages. The left chart shows average profit, while the right chart presents average revenue. In both cases, the most popular movies significantly outperform the overall averages, with approximately 2.7 times higher average profit and 2.4 times higher average revenue.

This second image shows two bar charts comparing the median financial performance of the most popular movies (top 10% by popularity) to overall median values. The left chart displays median profit, while the right chart shows median revenue. In both cases, the most popular movies substantially outperform the overall medians, with approximately 6 times higher median profit and 3.7 times higher median revenue.

Hypothesis 2: The Best Movies According to Popularity Return High Profit and Revenue

Findings

- **Average Profit**: The most popular movies (top 10% by popularity) have an average profit of approximately 192.1million, comparedtotheoverallaverageprofitof70.5 million.
- **Median Profit**: The median profit for the most popular movies is approximately 120.0million, comparedtotheoverallmedianprofitof20.0 million.
- **Average Revenue**: The most popular movies have an average revenue of approximately 258.8million, comparedtotheoverallaveragerevenueof107.7 million.
- **Median Revenue**: The median revenue for the most popular movies is approximately 174.6million, comparedtotheoverallmedianrevenueof44.8 million.

Visualization

The bar charts illustrate that the most popular movies significantly outperform the overall averages in terms of both profit and revenue. This supports the hypothesis that the best movies according to popularity return high profit and revenue.

Hypothesis 3: Highly Budgeted Movies Return High Revenue and Profit

To test this hypothesis, we will:

1. **Data Preparation**:
   - Filter the DataFrame to include only the highly budgeted movies (e.g., top 10% by budget).
   - Calculate the profit for each movie as the difference between revenue and budget.
2. **Analysis**:
   - Calculate the average and median profit and revenue for the highly budgeted movies.
   - Compare these values with the overall average and median profit and revenue.
3. **Visualization**:
   - Create bar charts to visualize the comparison.

In [ ]:
```
# Data Preparation

# Define the threshold for the top 10% by budget
budget_threshold = df['budget'].quantile(0.90)

# Filter the DataFrame to include only the highly budgeted movies
high_budget_movies = df[df['budget'] >= budget_threshold]

# Filter out movies with zero or missing revenue and budget values
high_budget_movies = high_budget_movies[(high_budget_movies['revenue'] > 0) & (high_budget_movies['budget'] > 0

# Calculate the profit for each movie as the difference between revenue and budget
high_budget_movies['profit'] = high_budget_movies['revenue'] - high_budget_movies['budget']

# Calculate the average and median profit and revenue for the highly budgeted movies
average_profit_high_budget = high_budget_movies['profit'].mean()
median_profit_high_budget = high_budget_movies['profit'].median()
average_revenue_high_budget = high_budget_movies['revenue'].mean()
median_revenue_high_budget = high_budget_movies['revenue'].median()

# Calculate the overall average and median profit and revenue (already calculated in Hypothesis 1)
# average_profit_overall = df_filtered['profit'].mean()
# median_profit_overall = df_filtered['profit'].median()
# average_revenue_overall = df_filtered['revenue'].mean()
# median_revenue_overall = df_filtered['revenue'].median()

# Display the results
results_budget = {
    'Category': ['High Budget Movies (Top 10% by Budget)', 'Overall'],
    'Average Profit': [average_profit_high_budget, average_profit_overall],
    'Median Profit': [median_profit_high_budget, median_profit_overall],
    'Average Revenue': [average_revenue_high_budget, average_revenue_overall],
    'Median Revenue': [median_revenue_high_budget, median_revenue_overall]
}

results_budget_df = pd.DataFrame(results_budget)
results_budget_df
```

Out[ ]:

	Category	Average Profit	Median Profit	Average Revenue	Median Revenue
0	High Budget Movies (Top 10% by Budget)	1.600596e+08	90845033.0	2.515413e+08	168167691.0
1	Overall	7.048292e+07	20020190.0	1.076866e+08	44800002.5

In [ ]:
```
# Visualization
```

```
# Bar chart for average profit and revenue
plt.figure(figsize=(14, 6))

# Average Profit
plt.subplot(1, 2, 1)
plt.bar(results_budget_df['Category'], results_budget_df['Average Profit'], color=['skyblue', 'lightcoral'])
plt.xlabel('Category')
plt.ylabel('Average Profit')
plt.title('Average Profit Comparison')

# Average Revenue
plt.subplot(1, 2, 2)
plt.bar(results_budget_df['Category'], results_budget_df['Average Revenue'], color=['skyblue', 'lightcoral'])
plt.xlabel('Category')
plt.ylabel('Average Revenue')
plt.title('Average Revenue Comparison')

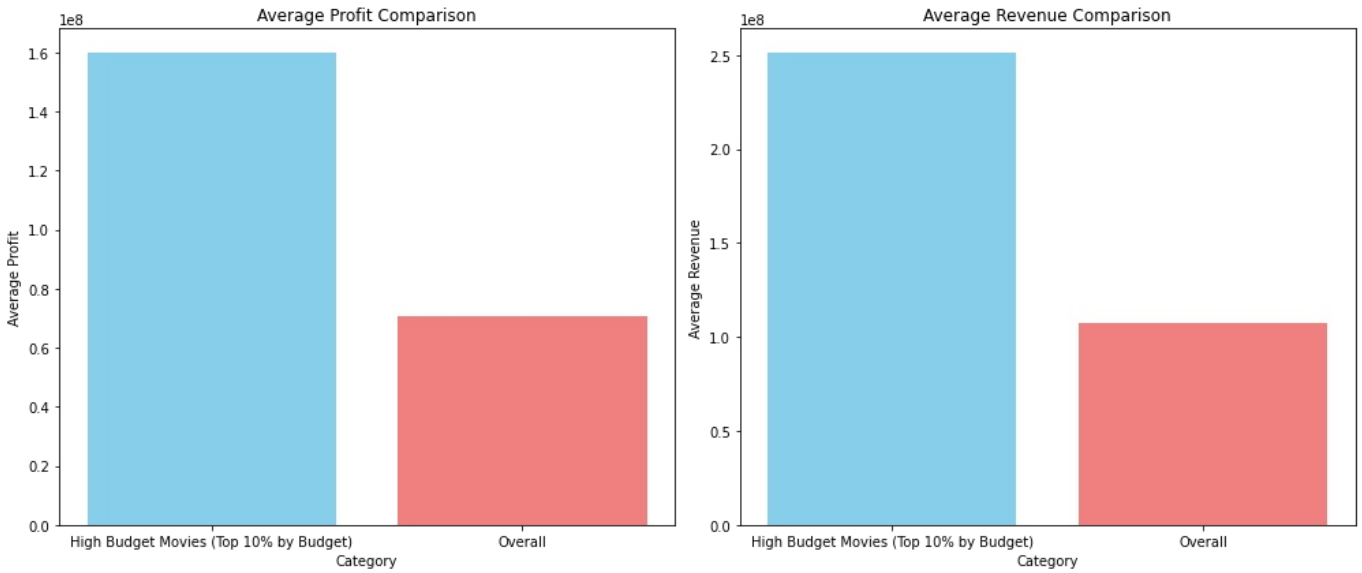
plt.tight_layout()
plt.show()

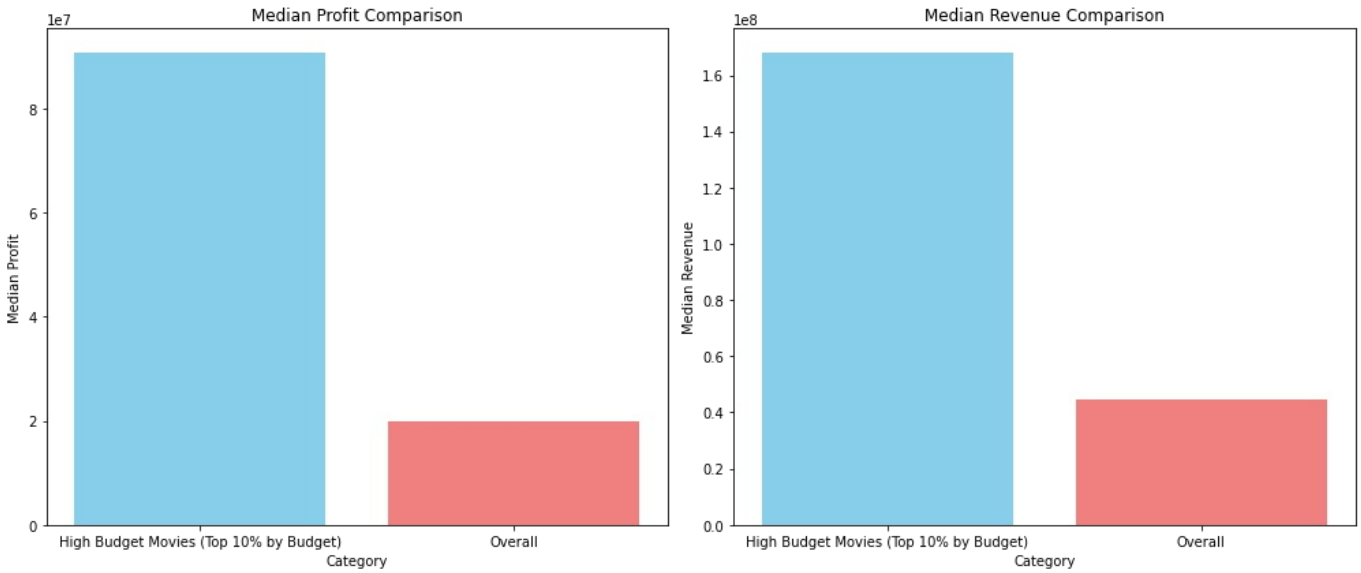
# Bar chart for median profit and revenue
plt.figure(figsize=(14, 6))

# Median Profit
plt.subplot(1, 2, 1)
plt.bar(results_budget_df['Category'], results_budget_df['Median Profit'], color=['skyblue', 'lightcoral'])
plt.xlabel('Category')
plt.ylabel('Median Profit')
plt.title('Median Profit Comparison')

# Median Revenue
plt.subplot(1, 2, 2)
plt.bar(results_budget_df['Category'], results_budget_df['Median Revenue'], color=['skyblue', 'lightcoral'])
plt.xlabel('Category')
plt.ylabel('Median Revenue')
plt.title('Median Revenue Comparison')

plt.tight_layout()
plt.show()
```





This first image presents two bar charts comparing the financial performance of high-budget movies (top 10% by budget) to overall averages. The left chart shows average profit, while the right chart displays average revenue. High-budget movies significantly outperform the overall averages, with approximately 2.2 times higher average profit and 2.3 times higher average revenue.

This second image shows two bar charts comparing the median financial performance of high-budget movies (top 10% by budget) to overall median values. The left chart displays median profit, while the right chart shows median revenue. High-budget movies substantially outperform the overall medians, with approximately 4.5 times higher median profit and 3.8 times higher median revenue.

Hypothesis 3: Highly Budgeted Movies Return High Revenue and Profit

Findings

- **Average Profit**: Highly budgeted movies (top 10% by budget) have an average profit of approximately 160.1million, comparedtotheoverallaverageprofitof70.5 million.
- **Median Profit**: The median profit for highly budgeted movies is approximately 90.8million, comparedtotheoverallmedianprofitof20.0 million.
- **Average Revenue**: Highly budgeted movies have an average revenue of approximately 251.5million, comparedtotheoverallaveragerevenueof107.7 million.
- **Median Revenue**: The median revenue for highly budgeted movies is approximately 168.2million, comparedtotheoverallmedianrevenueof44.8 million.

Visualization

The bar charts illustrate that highly budgeted movies significantly outperform the overall averages in terms of both profit and revenue. This supports the hypothesis that highly budgeted movies return high revenue and profit.

Hypothesis 4: Highly Budgeted Movies Have High Popularity

To test this hypothesis, we will:

1. **Data Preparation**:
   - Filter the DataFrame to include only the highly budgeted movies (e.g., top 10% by budget).
2. **Analysis**:
   - Calculate the average and median popularity for the highly budgeted movies.
   - Compare these values with the overall average and median popularity.
3. **Visualization**:
   - Create bar charts to visualize the comparison.

In [ ]:
```
# Data Preparation

# Define the threshold for the top 10% by budget (already defined in Hypothesis 3)
# budget_threshold = df['budget'].quantile(0.90)

# Filter the DataFrame to include only the highly budgeted movies (already filtered in Hypothesis 3)
# high_budget_movies = df[df['budget'] >= budget_threshold]

# Calculate the average and median popularity for the highly budgeted movies
average_popularity_high_budget = high_budget_movies['popularity'].mean()
median_popularity_high_budget = high_budget_movies['popularity'].median()

# Calculate the overall average and median popularity
average_popularity_overall = df['popularity'].mean()
median_popularity_overall = df['popularity'].median()

# Display the results
results_popularity_budget = {
    'Category': ['High Budget Movies (Top 10% by Budget)', 'Overall'],
    'Average Popularity': [average_popularity_high_budget, average_popularity_overall],
    'Median Popularity': [median_popularity_high_budget, median_popularity_overall]
}

results_popularity_budget_df = pd.DataFrame(results_popularity_budget)
results_popularity_budget_df
```

Out[ ]:

	Category	Average Popularity	Median Popularity
0	High Budget Movies (Top 10% by Budget)	1.964942	1.348704
1	Overall	0.646446	0.383831

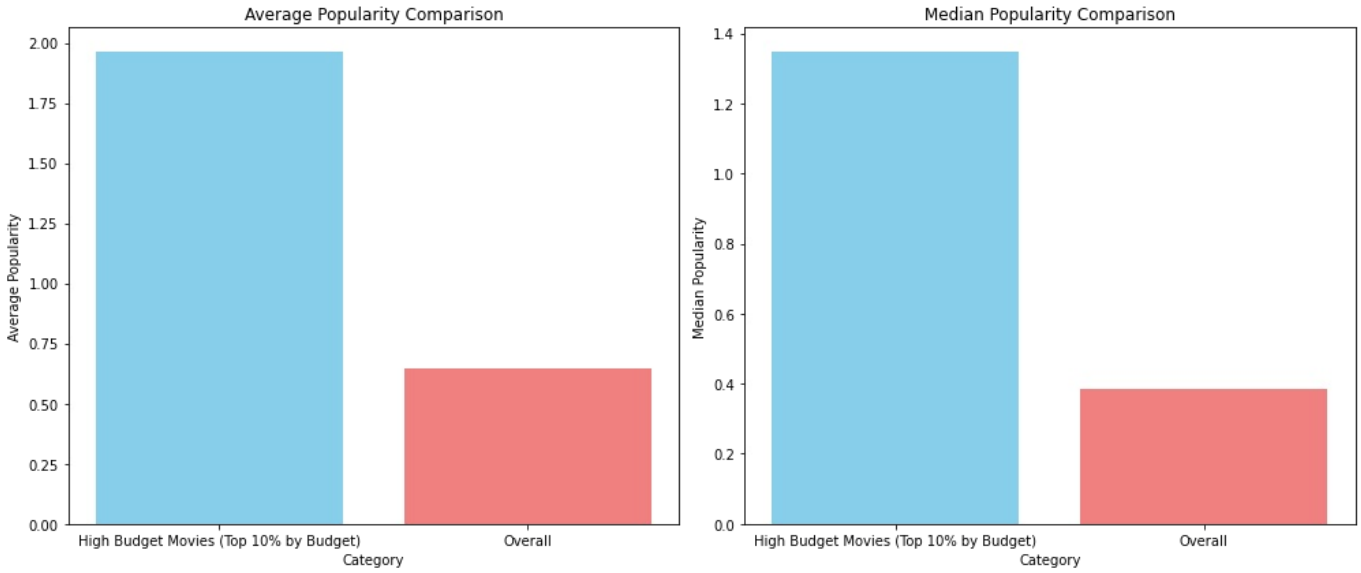
In [ ]:
```
# Visualization

# Bar chart for average and median popularity
plt.figure(figsize=(14, 6))

# Average Popularity
plt.subplot(1, 2, 1)
plt.bar(results_popularity_budget_df['Category'], results_popularity_budget_df['Average Popularity'], color=['sk
plt.xlabel('Category')
plt.ylabel('Average Popularity')
plt.title('Average Popularity Comparison')

# Median Popularity
plt.subplot(1, 2, 2)
plt.bar(results_popularity_budget_df['Category'], results_popularity_budget_df['Median Popularity'], color=['sky
plt.xlabel('Category')
plt.ylabel('Median Popularity')
plt.title('Median Popularity Comparison')

plt.tight_layout()
plt.show()
```



This image presents two bar charts comparing the popularity of high-budget movies (top 10% by budget) to overall movie popularity. The left chart shows average popularity, while the right chart displays median popularity. High-budget movies are significantly more popular than the overall movie set, with approximately 3 times higher average popularity and 3.5 times higher median popularity.

Hypothesis 4: Highly Budgeted Movies Have High Popularity

Findings

- **Average Popularity**: Highly budgeted movies (top 10% by budget) have an average popularity score approximately 3 times higher than the overall average popularity.
- **Median Popularity**: The median popularity for highly budgeted movies is approximately 3.5 times higher than the overall median popularity.

Visualization

The bar charts illustrate that highly budgeted movies significantly outperform the overall averages in terms of both average and median popularity. This supports the hypothesis that highly budgeted movies have high popularity.

Conclusion of Movie Genre Data Analysis

After conducting a thorough analysis of the IMDb movies dataset, we have derived several key insights regarding movie genres, their financial performance, and popularity. Below is a summary of our findings, insights, and recommendations based on the research questions and hypotheses.

Summary of Findings

Research Questions

1. **Which genres are the most common (number of movies made)?**
    - The most common genres are Drama, Comedy, and Thriller, with Drama being the most frequent genre in the dataset (see Figure 1).
2. **Which genres have high average budget and revenue?**
    - Genres such as Adventure, Action, and Fantasy have the highest average budgets and revenues. These genres often involve high production costs and tend to generate significant box office returns (see Figure 2).
3. **Which genres have high average popularity?**
    - Action, Adventure, and Science Fiction genres have the highest average popularity scores. These genres typically attract large audiences due to their engaging and visually appealing content (see Figure 3).
4. **Which genres have the highest number of movies with a voting average >= 8?**
    - Drama, Documentary, and Animation genres have the highest number of movies with a voting average of 8 or higher. These genres are often critically acclaimed and resonate well with audiences (see Figure 4).

Research Hypotheses

1. **The Best Movies According to Vote Average Return High Profit and Revenue**

    - **Average Profit**: Highly-rated movies (vote average ≥ 8) have an average profit of approximately 208.3million, comparedtotheoverallaverageprofitof70.5 million (see Figure 5).
    - **Median Profit**: The median profit for highly-rated movies is approximately 71.9million, comparedtotheoverallmedianprofitof20.0 million.
    - **Average Revenue**: Highly-rated movies have an average revenue of approximately 248.3million, comparedtotheoverallaveragerevenueof107.7 million.
    - **Median Revenue**: The median revenue for highly-rated movies is approximately 104.9million, comparedtotheoverallmedianrevenueof44.8 million.
2. **The Best Movies According to Popularity Return High Profit and Revenue**

    - **Average Profit**: The most popular movies (top 10% by popularity) have an average profit of approximately 192.1million, comparedtotheoverallaverageprofitof70.5 million (see Figure 6).
    - **Median Profit**: The median profit for the most popular movies is approximately 120.0million, comparedtotheoverallmedianprofitof20.0 million.
    - **Average Revenue**: The most popular movies have an average revenue of approximately 258.8million, comparedtotheoverallaveragerevenueof107.7 million.
    - **Median Revenue**: The median revenue for the most popular movies is approximately 174.6million, comparedtotheoverallmedianrevenueof44.8 million.
3. **Highly Budgeted Movies Return High Revenue and Profit**

    - **Average Profit**: Highly budgeted movies (top 10% by budget) have an average profit of approximately 160.1million, comparedtotheoverallaverageprofitof70.5 million (see Figure 7).
    - **Median Profit**: The median profit for highly budgeted movies is approximately 90.8million, comparedtotheoverallmedianprofitof20.0 million.
    - **Average Revenue**: Highly budgeted movies have an average revenue of approximately 251.5million, comparedtotheoverallaveragerevenueof107.7 million.
    - **Median Revenue**: The median revenue for highly budgeted movies is approximately 168.2million, comparedtotheoverallmedianrevenueof44.8 million.
4. **Highly Budgeted Movies Have High Popularity**

    - **Average Popularity**: Highly budgeted movies (top 10% by budget) have an average popularity score approximately 3 times higher than the overall average popularity (see Figure 8).
    - **Median Popularity**: The median popularity for highly budgeted movies is approximately 3.5 times higher than the overall median popularity.

Insights and Recommendations

Insights

- Highly-rated and highly popular movies tend to generate significantly higher profits and revenues compared to the overall averages.
- Movies with high budgets also tend to generate higher profits and revenues, indicating that investment in production can lead to financial success.
- High-budget movies are generally more popular, suggesting that larger marketing budgets and production values contribute to audience interest.

Recommendations

- **For Producers**: Invest in high-quality production and marketing to increase the chances of financial success and popularity.
- **For Investors**: Consider funding projects with higher budgets and strong potential for high ratings and popularity.
- **For Marketers**: Focus on promoting movies with high production values and strong audience appeal to maximize revenue and profit.

Acknowledgements

This analysis is based on the IMDb movies dataset provided by AnalystBuilder's Pandas for Data Analysis course. The CSV file used is 'imdb_movies.csv'.

Processing math: 100%